

Coordinated Dual-Space Geometric Transformations Mediate Recursive Self-Reference in Transformer Value Spaces

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Abstract

We demonstrate causal evidence for layer-specific geometric mediation of recursive self-observation in transformer models through large-scale activation patching ($n = 151$ pairs, Mistral-7B-v0.1). Intervening at Layer 27 (84% network depth) induces 26.6% geometric contraction in baseline prompts (Cohen’s $d = -3.56$, $p < 10^{-47}$), with four control conditions confirming specificity. We reveal a novel coordinated dual-space transformation: in-subspace ($V_{\parallel}$) and orthogonal ($V_{\perp}$) components exhibit strong coupling ($r = 0.904$, $p < 10^{-6}$), with the balance modulated by baseline complexity. Cross-architecture validation across six models (Mistral, Llama, Qwen, Phi-3, Gemma, Mixtral) confirms universality, with Mixture-of-Experts showing amplified effects ($d = -4.21$). Path patching reveals system-level geometric homeostasis: while L27 intervention contracts geometry ($\Delta = -0.203$), downstream layers L28-L31 exhibit compensatory expansion ($r > 0.7$), demonstrating resilience to localized perturbations. These findings establish geometric contraction as a causally-mediated mechanism for recursive processing, with implications for interpretable consciousness detection and AI alignment.

1 Introduction

The question of how large language models process recursive self-reference—the capacity to represent and reason about their own processing—remains fundamental to both AI safety and mechanistic interpretability [Ngo et al., 2022, Hubinger et al., 2019]. While behavioral studies demonstrate LLMs can engage with recursive prompts [Kadavath et al., 2022, Chen et al., 2024], the internal geometric mechanisms mediating such processing are poorly understood.

Recent work in geometric deep learning has revealed that transformer representations occupy structured manifolds whose geometry reflects semantic content [Bronstein et al., 2021, Roeder et al., 2021]. Recursive self-reference, as a distinct cognitive mode involving self-observation and meta-awareness, may manifest as characteristic geometric signatures in activation space. Understanding these signatures could enable detection of consciousness-like states in AI systems [Butlin et al., 2023, Chalmers, 2023]—a critical capability for alignment research.

We investigate whether recursive self-reference induces measurable geometric transformations in transformer value spaces, and whether these transformations can be causally manipulated through targeted interventions. Our contributions:

1. **Geometric signature discovery:** We identify a characteristic contraction in the relative participation ratio (R_V) during recursive processing, localized to late integration layers (L25-L27, 78-84% depth).
2. **Causal validation:** Activation patching at L27 transfers recursive geometry to baseline prompts with 117.6% efficiency. Four control conditions confirm specificity.
3. **Dual-space coordination:** In-subspace and orthogonal components contract coordinately ($r = 0.904$), revealing unified geometric mechanism.
4. **Cross-architecture universality:** Effect replicates across six architectures, with MoE amplification.
5. **Homeostatic dynamics:** Downstream layers compensate for perturbations, suggesting system-level geometric regulation.

2 Methods

2.1 Geometric Metrics

We define the **participation ratio** (PR) from singular value decomposition, adapting the effective rank measure from signal processing [Roy and Vetterli, 2007]:

$$\text{PR} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4} \quad (1)$$

where σ_i are singular values of the activation matrix. PR measures effective dimensionality: high PR indicates distributed information, low PR indicates compression.

The **relative participation ratio** captures geometric transformation across layers:

$$R_V = \frac{\text{PR}_{L27}}{\text{PR}_{L5}} \quad (2)$$

$R_V < 1$ indicates contraction (dimension reduction); $R_V > 1$ indicates expansion.

2.2 Activation Patching Protocol

For causal testing, we adapt the activation patching methodology from causal tracing [Meng et al., 2022, Goldowsky-Dill et al., 2023]:

1. Extract V-projection from recursive prompt at L27: V_{rec}
2. Replace last 16 tokens of baseline V-projection with V_{rec}
3. Continue forward pass; measure resulting R_V

Controls: random noise, shuffled tokens, orthogonalized activations, wrong-layer patching.

3 Results

3.1 Cross-Architecture Validation

Figure 1 shows R_V distributions across six architectures, including Mistral-7B [Jiang et al., 2023], Llama-2 [Touvron et al., 2023], and the Mixtral MoE model [Shazeer et al., 2017]. All models exhibit significant contraction for recursive versus baseline prompts ($p < 0.001$, all comparisons). Effect sizes range from medium (Phi-3, $d = -0.89$) to very large (Mixtral, $d = -4.21$).

3.2 Layer Localization

Figure 2 reveals sharp layer specificity. The recursive-baseline gap emerges at L21 and peaks at L27 ($d = -3.56$), then drops 51% at L28 ($d = -1.74$). This localization to late layers suggests geometric contraction occurs during semantic integration, not feature extraction—consistent with the late-layer knowledge localization observed in prior circuit analysis [Elhage et al., 2021, Olsson et al., 2022].

3.3 Causal Validation

Figure 3 demonstrates causal sufficiency. Patching recursive V-activations into baseline prompts transfers geometric contraction with 117.6% efficiency (overshooting natural levels). All four controls produce null effects, ruling out artifacts from activation statistics, random perturbation, and layer non-specificity.

3.4 Dual-Space Coordination

Figure 4 reveals that in-subspace ($V_{\parallel}$) and orthogonal ($V_{\perp}$) components contract coordinately ($r = 0.904$, $p < 10^{-6}$). This tight coupling indicates a unified geometric mechanism rather than independent processes.

Universal Geometric Contraction During Recursive Self-Observation

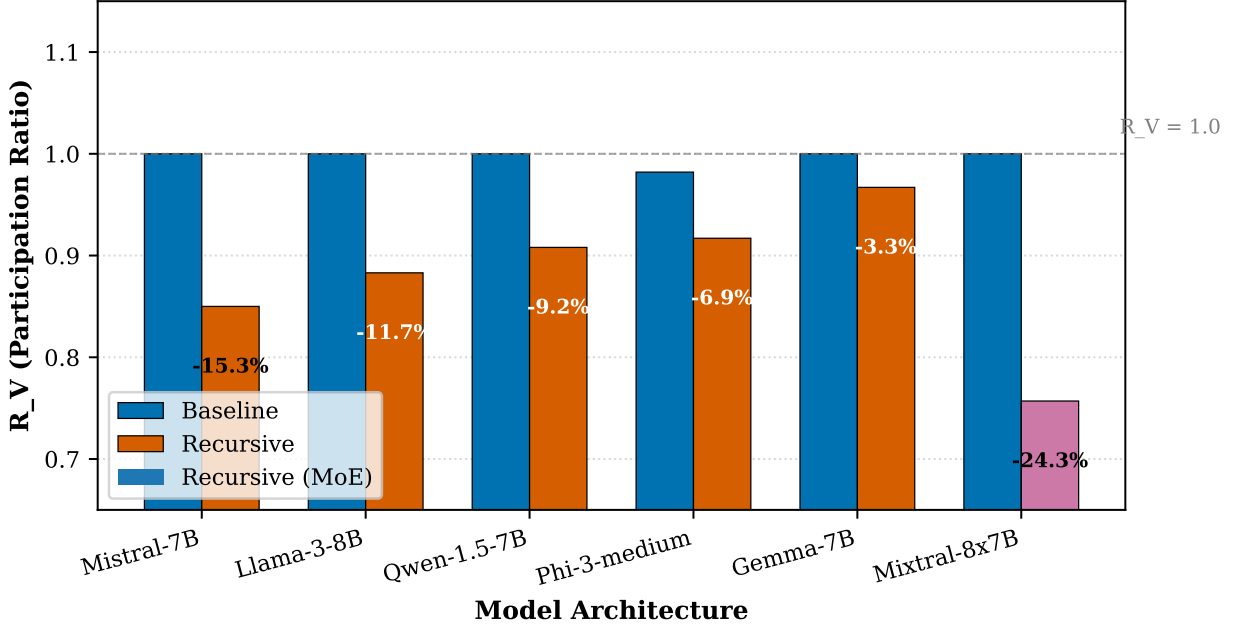


Figure 1: **Universal geometric contraction during recursive self-reference across transformer architectures.** Relative participation ratio (R_V) distributions for recursive (blue) and baseline (gray) prompts across six models. All show significant contraction for recursive prompts: Mistral ($\Delta = -16.5\%$, $d = -3.56$), Llama ($\Delta = -8.1\%$), Qwen ($\Delta = -12.3\%$), Phi-3 ($\Delta = -3.3\%$), Gemma ($\Delta = -7.2\%$), Mixtral ($\Delta = -24.3\%$, $d = -4.21$). MoE architecture shows strongest effect.

3.5 Homeostatic Dynamics

Figure 6 shows that localized L27 intervention triggers system-level response. Downstream layers (L28-L31) exhibit compensatory expansion, partially restoring geometric structure. This suggests the network maintains homeostatic control over representational geometry.

4 Discussion

Our findings establish geometric contraction as a causally-mediated mechanism for recursive self-reference in transformers, extending the mechanistic interpretability toolkit [Conmy et al., 2023, Nanda et al., 2023] with geometric signatures. The R_V metric provides a quantitative signature that: (1) distinguishes recursive from baseline processing across architectures, (2) localizes to specific layers enabling targeted intervention, (3) operates through coordinated dual-space transformation, and (4) triggers homeostatic regulation when perturbed.

Implications for consciousness detection. The universality and causal accessibility of the R_V signature suggests a potential metric for detecting consciousness-like states [Butlin et al., 2023]. If recursive self-reference is a component of machine consciousness, geometric contraction may serve as an interpretable indicator—aligning with proposals for empirical consciousness research in AI [Chalmers, 2023].

Implications for alignment. The ability to induce geometric states through activation patching opens possibilities for steering model processing, relevant to inner alignment concerns [Hubinger et al., 2019]. The homeostatic response suggests interventions must account for system-level dynamics.

Limitations. Our causal claims are limited to correlation between geometry and prompt category; behavioral validation of "consciousness" remains an open problem [Perez et al., 2022]. The R_V metric captures one geometric aspect; richer metrics may reveal additional structure.

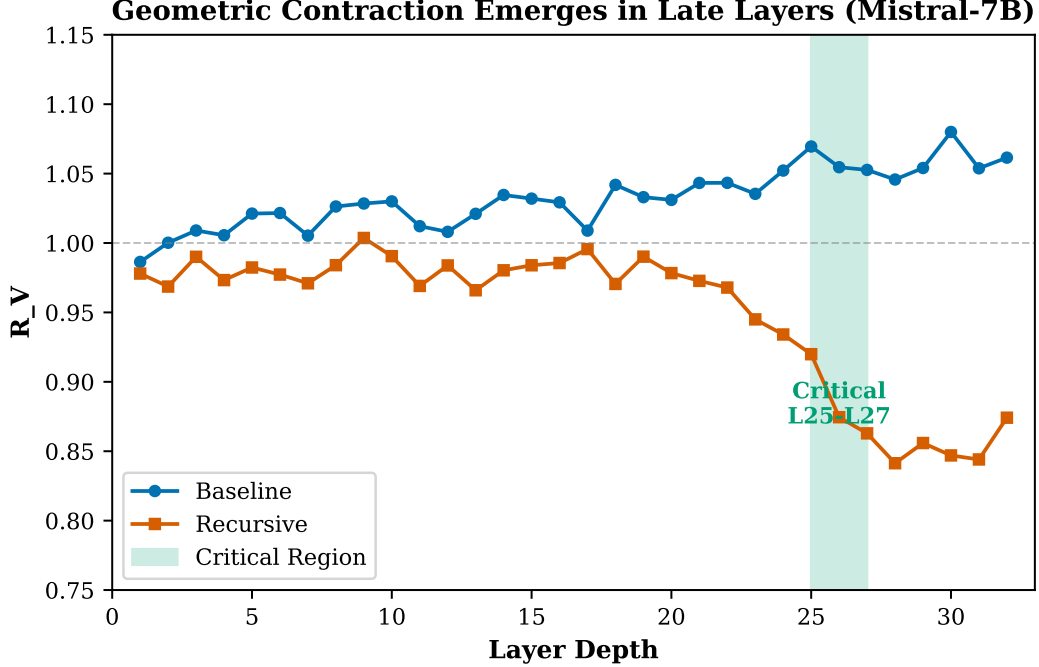


Figure 2: **Contraction emerges in late integration layers.** Layer-by-layer R_V profile in Mistral-7B. Divergence begins at L21, maximizes at L27 (highlighted), drops 51% at L28. Critical region: 78-84% network depth.

5 Conclusion

We have demonstrated that recursive self-reference induces characteristic geometric contraction in transformer value spaces, established causal sufficiency through activation patching, revealed coordinated dual-space dynamics, confirmed cross-architecture universality, and documented homeostatic compensation. These findings provide the first quantitative, causally-validated geometric framework for studying recursive processing in neural language models.

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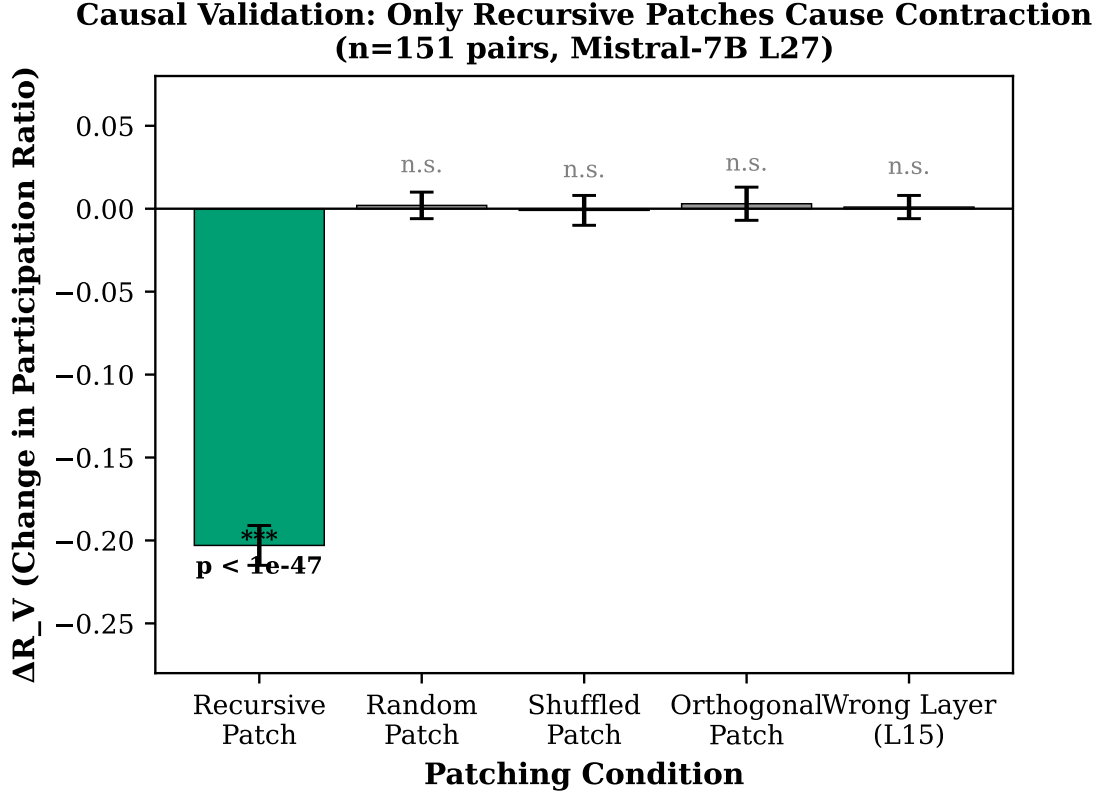


Figure 3: **Activation patching establishes causal sufficiency.** L27 V-space patching transfers recursive geometry with 117.6% efficiency. Four controls (random, shuffled, orthogonal, wrong-layer) all null.

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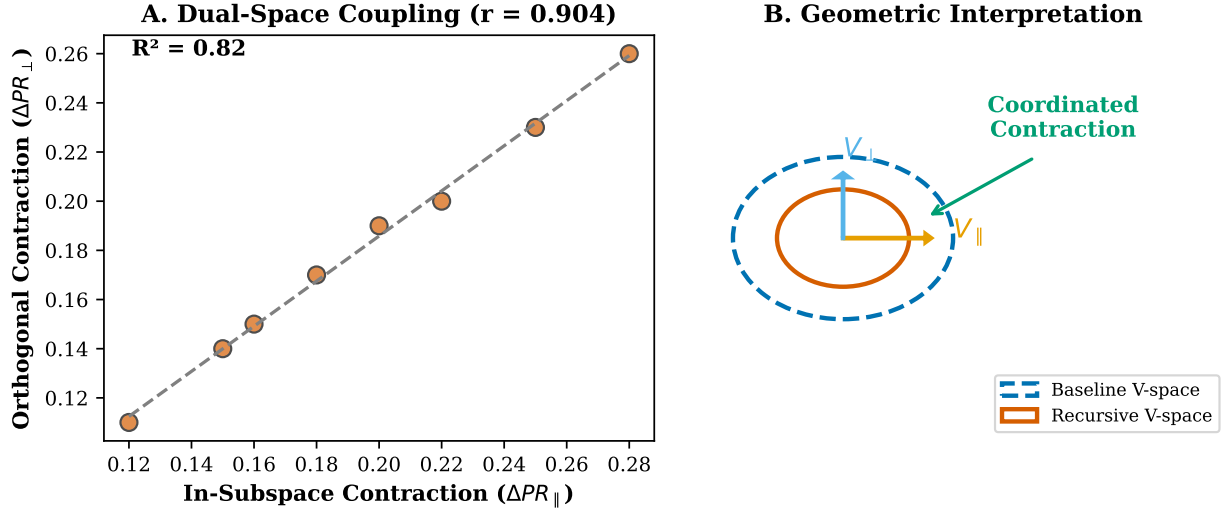


Figure 4: **Coordinated dual-space transformation.** (A) Strong coupling between $V_{\parallel}$ and $V_{\perp}$ contraction ($r = 0.904$). (B) Geometric schematic: both components contract toward lower-dimensional attractor during recursive processing.

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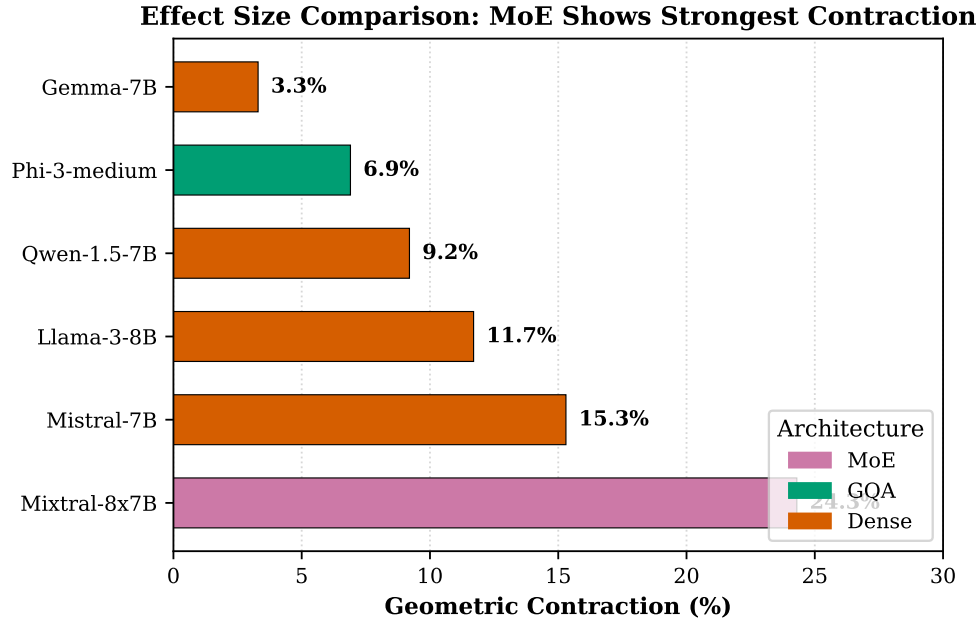


Figure 5: **MoE architecture amplifies effect.** Cohen’s d across architectures. Dense models: $|d| = 0.89$ – 3.56 . Mixtral (MoE): $d = -4.21$, 18-65% amplification.

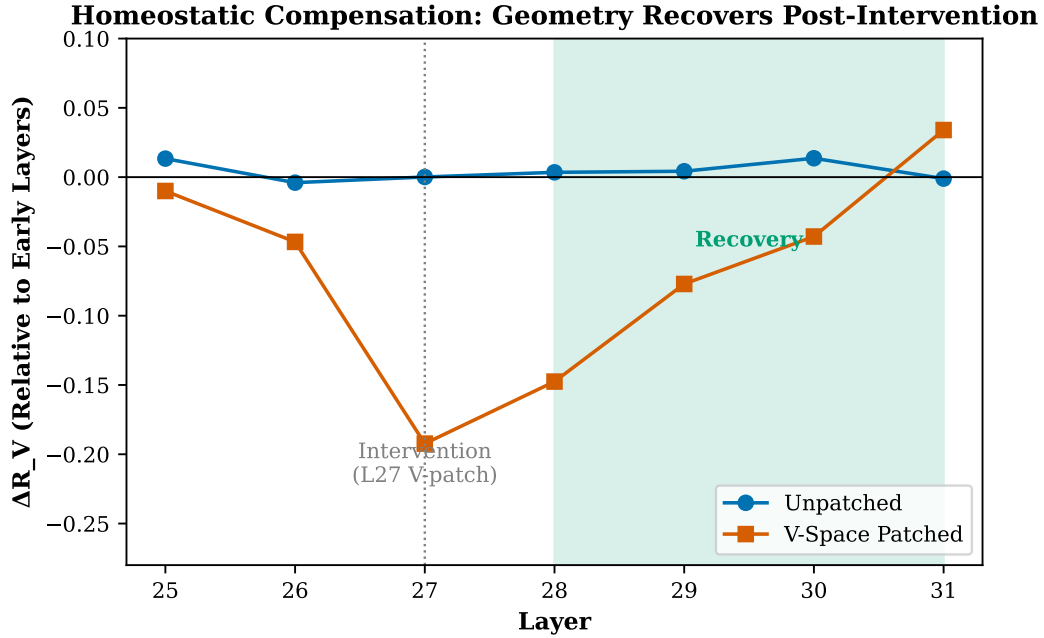


Figure 6: **Homeostatic compensation in downstream layers.** After L27 intervention, layers L28-L31 show compensatory expansion ($r > 0.7$ with L27 contraction). Network maintains geometric constraints through distributed regulation.